Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/17/2019
PI Name:	Avedisian, C. Thomas Ph.D.		
Project Title:	Use of FLEX Data to Enhance Video Image Processing for Studying Combustion of Soot-Producing Jet Fuel Droplets and Their Blends with Bio-Derived Fuels		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cta2@cornell.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	607-592-7915
Organization Name:	Cornell University		
PI Address 1:	Department of Mechanical and Aerospace Engineering		
PI Address 2:	Upson and Grumman Halls		
PI Web Page:			
City:	Ithaca	**State:**	NY
Zip Code:	14853-7501	**Congressional District:**	23
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015-16 Physical Sciences NNH15ZTT001N-15PSI-C: Use of the NASA Physical Sciences Informatics System – Appendix C
Start Date:	01/15/2018	**End Date:**	01/14/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hicks, Michael C.	**Contact Phone:**	216-433-6576
Contact Email:	Michael.C.Hicks@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Reeves, Anthony Ph.D. (Cornell University)		
Grant/Contract No.:	80NSSC18K0480		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	One of the most important challenges of our time is to mitigate the harmful impacts of particulate emissions (soot) from combustion engines on global climate change, and to reduce the consumption of liquid petroleum fuels that power transportation systems. A promising concept is to mix petroleum fuels with bio-derived fuels. An evaluation of the performance benefits rests on diagnostics to unravel the complex relationships between soot formation and fuel properties. Under some conditions the presence of soot itself in the combustion field can intrude on this effort. Measurements that rely on digital video imaging of the fuel burning process, such as employed in the FLame Extinguishment eXperiment (FLEX), are vulnerable to this concern. FLEX was developed to understand mechanisms that control flame extinction as it relates to cabin fire safety aboard manned spaceflight, though the broad methodologies are more widely applicable to combustion of fuel systems in combustion engines. The configuration of isolated droplets burning with near spherical symmetry was employed to elucidate the mechanisms involved. The main diagnostic was digital video imaging. From this information, quantitative measurements of the droplet and flame boundaries were made using suitable computer-based data extraction algorithms that revealed the influence of ambient gas conditions and liquid fuel properties on flame extinction. For some of the fuels employed in FLEX, soot formation was evident (i.e., n-heptane droplets burning in air), which obscured part of the droplet and flame boundary that can compromise data accuracy. The success of the FLEX program assumes the existence of efficient and accurate computer-based algorithms that can overcome the obscuration influence of soot on droplets, are able to track droplets if they move, and can account for deviations from sphericity due to ignition disturbances. However, this is not the case. Commercial image analysis packages are limited in their ability to deal with the presence of soot in the imaging field without significant user input. Failure to address this problem impacts not only deriving full benefit from the FLEX experiments already carried out, but also follow-on studies which would utilize digital video imaging to study droplet combustion dynamics of sooting fuels used in commercial transportation systems. In this proposal we address this problem be developing a diameter estimation refinement algorithm that employs concepts that automate droplet and flame diameter measurements from video images in the presence of soot. The algorithm will be used to re-analyze certain FLEX data on the Physical Sciences Informatics (PSI) system -- results from a proof-of-concept program are included in the proposal for illustration -- and to extract quantitative data from new proposed experiments to study the influence of mixture fraction on blends of bio-derived fuels with an aviation fuel: hydroprocessed renewable jet fuel derived from algae with Jet-A (algal feedstocks have potential for larger yields compared to outer sources) for which soot formation is expected. The outcome will be a robust capability to process digital video images that efficiently and precisely extracts measurements from digital images even in the presence of soot aggregates. The results -- data and software -- will be freely available on various outlets as discussed in the proposal. The project includes a team with expertise in developing algorithms for image analysis (Prof. Anthony Reeves of Cornell's Electrical and Computer Engineering Department) and experience in droplet combustion experimentation (Prof. C. Thomas Avedisian of Cornell's Mechanical and Aerospace Engineering (MAE) Department). The project team has supplies of the algae-derived and jet fuel systems that will be studied (Terravia, Inc; and the Air Force).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	One of the most important challenges of our time is to mitigate the harmful impacts of particulate emissions (soot) from combustion engines on global climate change, and to reduce the consumption of liquid petroleum fuels that power transportation systems. A promising concept is to mix petroleum fuels with bio-derived fuels. An evaluation of the performance benefits rests on diagnostics to unravel the complex relationships between soot formation and fuel properties. Under some conditions the presence of soot itself in the combustion field can intrude on this effort. Measurements that rely on digital video imaging of the fuel burning process, such as employed in the FLame Extinguishment eXperiment (FLEX), are vulnerable to this concern. FLEX was developed to understand mechanisms that control flame extinction as it relates to cabin fire safety aboard manned spaceflight, though the broad methodologies are more widely applicable to combustion of fuel systems in combustion engines. The configuration of isolated droplets burning with near spherical symmetry was employed to elucidate the mechanisms involved. The main diagnostic was digital video imaging. The primary goal of this research is to develop precision computer algorithms that can measure the diameter of near-spherical fuel droplets in each video image. In previous work general purpose semi-automated measurement tools have been used which suffer to main drawbacks: First the standard algorithms often fail in the presence of soot and second there is no error confidence interval determined for any measurement so the precision of these measurements is unknown. At the outcome we will have a general purpose fully automated image evaluation system that will provide the precision of the diameter measurement for each image. This tool may be applied to any situation where a precise shape measurement of a moving object in a video image. A second goal is to explore mixtures of recent bio-derived fuels that were not considered in the FLEX2 experiments. These are conducted in a drop tower experimental system at Cornell. This work will provide insight in the advantages of the new mixtures. While these are currently analyzed using traditional semi-automated methods we plan to reanalyze the data once the automated precision methods are validated for this data.
	In the first phase of this research we developed a general tracking and measuring system for any image sequence of a moving circle. This involved a tracking stage that identified the location of the circle in each image and a precision diameter measuring stage. The tracking system could address very large motions of the droplet between images. For measurement the system employs an iterative ray casting method that measures each radial ray to subpixel accuracy in a directed gradient image. In the second phase, we focused on the soot problem with backlight droplet image videos. For this task, changes between images were small and the tracking stage was not required. However, there were a number of different soot types that required different algorithm modifications. The different soot types are powder soot, in which particles smaller than a pixel darken the background, granular soot in which particles may be several pixels in size and cause a positive bias to the diameter measurement, and occluding soot that is large opaque consolidated masses of soot that, in extreme cases, can completely cover the droplet and make measurement impossible. For the powdered soot no explicit modifications to the algorithm are needed since measurement is performed a sub-pixel accuracy on a directed gradient image. This approach has shown to give very good results when traditional image-intensity based semiautomated methods fail. For the granular soot there is a measurement bias associated with the

Task Progress:	radial measurements going to the boundary of soot granules that connect to the the boundary of the droplet. We are exploring two methods to correct these errors--first a set of outlier radial measures are discounted since they have lower reliability; second a modification is made to the loss function of the Mean Squared Error (MSE) optimization algorithm used to refine the diameter measurement. For the occlusion there are situations that we have recently found where the whole droplet is occluded and no measurement is possible. To deal with situation we have developed a linear predictive tracking algorithm that carefully reviews a large location and size search region following the detection of a full occlusion event. In this way the algorithm recaptures the droplet when it eventfully comes out from behind the occluding soot cloud. No measurements are recorded in a region where there is full occlusion. For each image measurement, 7 different image quality measurements are made in order to estimate measurement error; these include standard deviation of the radial boundary measurements, mean signal of the boundary, and the number of discarded radii due to there being no gradient peak or being a significant outlier. In the next phase of this research we will use the image quality measures to for a model for predicting measurement error by the use of Monte Carlo simulations. To date we have evaluated this algorithm on 125 video image sequences for all the FLEX2 involving the following fuel mixtures: Propylbenzene100, Decane25 Propylbenzene75, Decane50 Propylbenzene50, Decane75 Propylbenzene25, and Heptane50 Toluene50. Using our phase 1 method we successfully measured 57% of these sequences. By using the advanced measurement methods to manage soot we have increased this to 88% success. With revisions to our prediction method we plan to be able to successfully measure all the image sequences. In a second study based on a local drop tower for data, we report the results of a fundamental study of hydroprocessed renewable jet fuel derived from heterotrophic algae (AHRJ) burning under conditions that promote one-dimensional transport dynamics and near spherically symmetric burning conditions. The results are compared to Jet-A and three AHRJ/Jet-A blends containing 25%, 50%, and 75% AHRJ by volume. The results show that AHRJ droplets burn slightly faster than Jet-A while adding AHRJ to Jet-A reduces the sooting propensity. These trends suggest the viability of blending Jet-A with AHRJ to fuel combustion engines.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Abstracts for Journals and Proceedings	Reeves AP, Xu Y, Avedisian CT, Hicks MC. "Combustion of N-decane and N-propylbenzene droplets: data analysis and effect of droplet diameter." Presented at 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Abstract ID #81. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Nov-2018